Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> For part (a), we need to calculate the heat required to raise the temperature of 1 liter of water from 30Â°C to 60Â°C. The specific heat of water is 1 cal/gÂ°C. The mass of 1 liter of water is approximately 1000 grams. The heat required is given by Q = mass * specific heat * change in temperature. Q = 1000 g * 1 cal/gÂ°C * (60Â°C - 30Â°C) = 30000 cal = 30 Kcal.

For part (b), we need to calculate the heat required to raise the temperature of 1 kg of aluminum from 30Â°C to 60Â°C. The specific heat of aluminum is 0.215 cal/gÂ°C. The mass of 1 kg of aluminum is 1000 grams. The heat required is given by Q = mass * specific heat * change in temperature. Q = 1000 g * 0.215 cal/gÂ°C * (60Â°C - 30Â°C) = 3225 cal = 7.77 Kcal.

Therefore, the answer is 30 Kcal, 7.77 Kcal.

</think>

<answer>30 Kcal, 7.77 Kcal</answer>